IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

MSc and EEE/EIE PART IV: MEng and ACGI

DISCRETE-EVENT SYSTEMS

Wednesday, 8 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : D. Angeli
Second Marker(s) : E. Kerrigan

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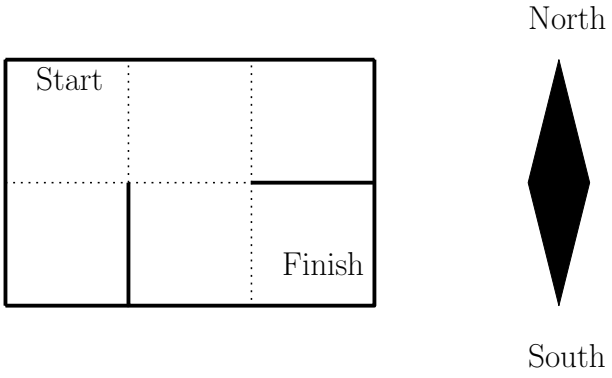


Figure 1.1 The labyrinth: layout and orientation

1. A robot moves within a "labyrinth" with 6 rooms (highlighted by dotted lines in Fig. 1.1), and oriented as shown by the north-south arrow. Thick lines denote walls that cannot be crossed, while dotted lines can be crossed without any difficulty. Each time the robot walks from one room to another an event is generated, of type e, w, n, s, for movement in the East, West, North or South directions, respectively. The robot starts in the North-West corner of the labyrinth (Start room) and needs to reach the South-East corner (Finish room).

   a) Define a finite deterministic automaton G_R to model the allowed movement of the robot within the rooms, list its events, and sketch its transition diagram. [5]

   b) Assume next that events n and w are unobservable. Replace these events with a single event m, which stands for mistake (as the goal is to reach the finish without ever needing to move north or west). Denote the new automaton by $\tilde{G}_R$. Design a labeling automaton G_L that monitors occurrence of m events. In particular, m events should trigger transition between an N state (indicating that m has never been triggered) and a Y state (indicating that m has occurred at least once). [2]

   c) Compute the parallel composition $G = \tilde{G}_R || G_L$. [4]

   d) Replace m events in G by the ε empty event, so as to achieve a non-deterministic automaton G_N. [1]

   e) Compute the observer automaton $\mathrm{Obs}(G_N)$. [5]

   f) Explain how $\mathrm{Obs}(G_N)$ can detect if an event of type m has been triggered by only observing events of type s and e. [3]

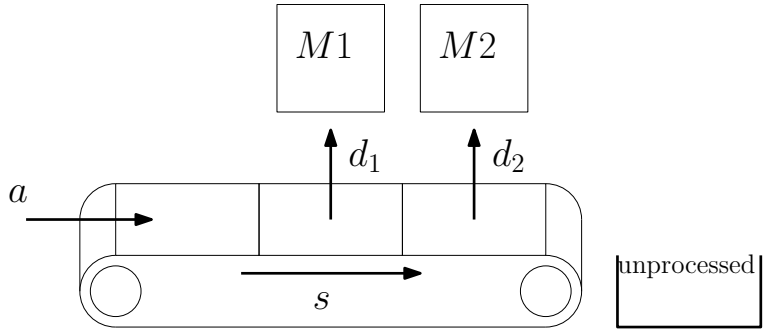


Figure 2.1 Conveyor belt and the machines layout

2. A conveyor belt, with 3 available slots, feeds 2 machines $M1$ and $M2$ respectively, as shown in Fig. 2.1. Items arrive on the belt at its leftmost slot (event a) provided the slot is available, and are carried towards the machines through discrete rotation movements of the belt (event s, shift). When an item is in front of Machine Mi, an event d_i can occur and the piece is then removed from the belt and transferred to the machine Mi where it will be processed in due time. If a shift event occurs when a piece is on the rightmost slot, then the item will drop into the box of unprocessed pieces.

   a) Model the conveyor belt, occupancy of its slots and the departure and arrival process of pieces through a finite deterministic automaton. [6]

   b) Consider the following informal specification: "all pieces need to be processed by either machine $M1$ or $M2$". Model this specification through an automaton G_S. [4]

   c) Assume that the events d_1 and d_2 are uncontrollable. Is the language $\mathscr{L}(G_S)$ controllable with respect to $E_{uc} = \{d_1, d_2\}$? If so sketch the corresponding supervisor. [4]

   d) Consider instead the case in which event s is uncontrollable. Is the language $\mathscr{L}(G_S)$ controllable with respect to $E_{uc} = \{s\}$? (Justify your answer). [2]

   e) Compute the maximal controllable sublanguage of $\mathscr{L}(G_S)$. [4]

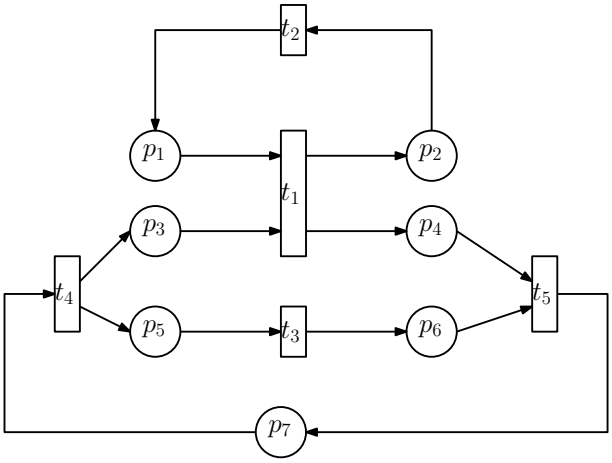


Figure 3.1 Petri Net of the assembly line

3. An assembly line can simulatneously process k items, for some positive integer k. The capacity is expressed through the initial marking in place p_7 of the Petri Net shown in Fig. 3.1, (provided all other activities have still to be initiated). When the assembling starts on a piece, 2 separate threads of work are simultaneously activated, by transition t_4. The first one, in turn can only proceed when a certain type of $3d$ printing units is available. The total number of such available units is expressed by the marking in place p_1. The number of busy printing units is instead represented by the marking in place p_2.

a) Consider the Petri Net N, shown in Fig. 3.1. Derive its *Pre*, *Post* and incidence matrix C, by ordering the places according to $p_1 \cdots p_7$ and transitions according to $t_1 \cdots t_5$. [5]

b) Compute the P-semiflow of minimal support of the network and discuss their physical interpretation, in the light of the description provided above. [4]

c) Show that the Petri Net is structurally weakly conservative. [2]

d) Compute the T-semiflow of minimal support. [3]

e) Consider the initial marking $M_0 = [1,0,0,0,0,0,1]$. Find the reachable set $\mathscr{R}(M_0)$ and sketch the associated transition diagram. [4]

f) Discuss, for the selection of M_0 considered in the previous item, the reversibility of the marked Petri Net N, M_0. [2]

4. A factory operates with 2 identical machines working in parallel. Each machine may transition to a faulty state according to a random exponentially distributed clock of rate λ_f. Once machines are in a faulty state, a technician operates on them to bring them back to their working state. They can, however, only work on a single machine at a time. Given that a range of different faults may occur, it has been verified that the time it takes to fix a single machine is also a random variable, distributed according to an exponential clock, of rate λ_r (restore phase).

a) Model the factory and the number of available operational machines as a Markov chain. Sketch its transition diagram and write its Kolmogorov equations, describing the evolution in time of the probability of being in each state. [6]

b) Is this Markov chain ergodic? (justify your answer). [2]

c) Assume that the factory's throughput is proportional to the number of operational machines at each point in time, i.e. equal to $2K$ units per second when 2 machines are operational, K units per second when 1 machine is operational and 0 units per second when all machines are faulty. Compute the asymptotic average factory throughput as a function of K and rates λ_f and λ_r. [5]

d) Show that the throughput can be improved by increasing λ_r. [2]

e) Assuming that the factory always starts out with 2 operational machines, compute the average time it takes to reach a state where all machines are faulty. [5]